Table 1. Characteristics of Included Implantable Neuromodulation Systems and Publicly Disclosed Digital Ecosystem Features.

System	Manufacturer	PMA #	Product Code	Indication	Implant Location	Neuromodulation Type	Sensing Capability (Y/N)	Remote Monitoring Capability (Y/N)	Remote Monitoring or Programming (Y/N)	Patient App (Y/N)	App on Patient's Own Phone (Y/N)	App on Company-Provided Device (Y/N)	Clinician Portal (Y/N)	Cloud Upload (Y/N)	Data Export Features (Y/N)	Patient App Name	Clinician Portal Name	Programmer Type (Dedicated vs Tablet)
Deep brain stimulation (DBS) - 3 systems																		
Boston Scientific Vercise Deep Brain Stimulation System	Boston Scientific	P150031	NHL	Parkinson's disease, essential tremor (DBS)	Brain	DBS	No	ND	ND	Yes	ND	Yes	ND	ND	Yes	Vercise DBS Controller	ND	Tablet
Medtronic Percept PC/RC	Medtronic	P960009	MHY	Parkinson's disease, essential tremor, epilepsy, dystonia and OCD under HDE (DBS)	Brain	DBS	No	ND	ND	Yes	No	Yes	ND	ND	Yes	Percept PC/RC Patient Programmer (Samsung smartphone)	ND	Tablet
Abbott Infinity DBS System	Abbott	P140009	MHY	Parkinson's disease, essential tremor (DBS)	Brain	DBS	No	ND	Yes	Yes	Yes	Yes	ND	Yes	Yes	NeuroSphere Patient Controller app	ND	Tablet
Spinal cord stimulation (SCS) - 6 systems																		
Medtronic Intellis	Medtronic	P840001	LGW	Chronic pain (SCS)	Spine	SCS	Yes	ND	ND	Yes	Yes	Yes	Yes	ND	Yes	MyStim RC	ND	Tablet
Boston Scientific (Nalu) Neurostimulation System	Boston Scientific (Nalu)	K233801	GZB	Chronic pain (SCS/PNS)	Spine/Peripheral nerve	SCS, Peripheral	No	ND	ND	Yes	Yes	Yes	ND	Yes	Yes	Nalu Remote Control app	ND	Tablet
Globus Medical (Nevro) Senza SCS System	Globus Medical (Nevro)	P130022	LGW	Chronic pain (SCS)	Spine	SCS	No	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	HFX app	HFX Cloud, Nevro healthcare provider portal	Dedicated
Saluda Medical Evoke SCS System	Saluda Medical	P190002	LGW	Chronic pain (SCS)	Spine	SCS	Yes	No	No	No	No	No	ND	ND	ND	N/A	ND	Tablet
Curonix Freedom SCS/PNS System	Curonix (formerly Stimwave)	K182720 (SCS), K233162 (PNS)	GZB, GZF	Chronic pain (SCS/PNS)	Spine/Peripheral nerve	SCS, Peripheral	No	No	No	No	No	No	No	No	ND	N/A	N/A	Tablet
Biotronik Prospera SCS System	BIOTRONIK NRO	P210037	LGW	Chronic pain (SCS)	Spine	SCS	No	Yes	Yes	Yes	ND	Yes	Yes	Yes	Yes	MyHomeStream (company smartphone)	Embrace One Provider Portal	Tablet
Vagus nerve stimulation (VNS) - 3 systems																		
LivaNova VNS Therapy for Epilepsy	LivaNova	P970003	LYJ	Refractory epilepsy	Chest/neck (vagus nerve)	VNS	Yes	ND	ND	ND	ND	No	ND	ND	Yes	ND	ND	Tablet
Mobia Medical Vivistim Paired VNS System	Mobia Medical (previously MicroTransponder)	P210007	QPY	Stroke rehabilitation	Chest/neck (vagus nerve)	VNS	No	No	No	No	No	No	No	No	Yes	N/A	N/A	Dedicated
SetPoint Medical SetPoint System	SetPoint Medical	P240039	SFJ	Rheumatoid arthritis	Neck (vagus nerve)	VNS	No	ND	ND	No	No	No	ND	Yes	Yes	N/A	ND	Tablet
Responsive neurostimulation (RNS) - 1 system																		
NeuroPace RNS System	NeuroPace	P100026	PFN	Epilepsy	Brain	RNS	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Tablet Remote Monitor	PDMS and nSight Platform	Tablet
Peripheral stimulation - 12 systems																		
Boston Scientific (Axonics) Sacral Neuromodulation System	Boston Scientific (Axonics)	P180046	EZW	Urinary/Bowel Dysfunction (SNM)	Pelvic (sacral nerve)	Peripheral	No	No	No	Yes	Yes	No	ND	Yes	Yes	myAxonics app	ND	Tablet
Bioness (Bioventus) StimRouter Neuromodulation System	Bioness (Bioventus)	K190047	GZF	Peripheral nerve pain	Peripheral nerve	Peripheral	No	No	No	Yes	Yes	ND	No	No	Yes	StimRouter Plus app	N/A	Tablet
CVRx BAROSTIM NEO System	CVRx	P180050	DSR	Heart failure	Neck (carotid sinus)	Peripheral	No	ND	ND	No	No	No	ND	ND	Yes	N/A	ND	Tablet
Inspire Medical Inspire V Upper Airway Stimulator	Inspire Medical Systems	P130008	MNQ	Obstructive sleep apnea	Chest/neck (hypoglossal nerve system)	Peripheral	Yes	ND	ND	Yes	Yes	No	Yes	Yes	Yes	Inspire Sleep app	Inspire SleepSync (Inspire Cloud)	Tablet
Mainstay Medical ReActiv8 Neurostimulation System	Mainstay Medical	P190021	QLK	Chronic low back pain	Lower back	Peripheral	No	No	No	No	No	No	No	No	Yes	N/A	N/A	Dedicated
Enterra Medical Enterra Therapy	Enterra Medical	HDE H990014	LNQ	Gastroparesis	Abdomen	Peripheral	No	No	No	No	No	No	No	No	ND	N/A	N/A	Dedicated
Medtronic InterStim	Medtronic	P970004	EZW	Urinary/Bowel Dysfunction (SNM)	Pelvic (sacral nerve)	Peripheral	No	ND	ND	Yes	Yes	Yes	ND	Yes	Yes	MyFuture app (patient phone), Medtronic smart programmer (company device)	ND	Dedicated
Neuros Medical Altius Nerve Stimulation System	Neuros Medical	P230020	SAZ	Post-amputation pain	Lower limb nerve	Peripheral	No	No	No	No	No	No	No	No	ND	N/A	N/A	Dedicated
Synapse Biomedical NeuRx Diaphragm Pacing System	Synapse Biomedical	P200018	OIR	Ventilator-dependent high spinal cord injury (diaphragm pacing)	Thorax/diaphragm	Peripheral	No	No	No	No	No	No	No	No	ND	N/A	N/A	Dedicated
Boston Scientific (Valencia) eCoin Peripheral Neurostimulator	Boston Scientific (Valencia Technologies)	P200036	QPT	Urinary urgency incontinence	Lower leg/ankle	Peripheral	No	No	No	No	No	No	No	No	ND	N/A	N/A	Dedicated
ZOLL remedē Phrenic Nerve Stimulator System	ZOLL (Respicardia)	P160039	PSR	Central sleep apnea	Chest (phrenic nerve)	Peripheral	Yes	No	No	No	No	No	No	No	ND	N/A	N/A	Tablet
Medtronic (SPR) SPRINT PNS System	SPR Therapeutics (now Medtronic)	K240974	SCN (earlier clearances: NHI)	Chronic and acute pain (60-day PNS)	Peripheral nerve (percutaneous lead)	Peripheral	No	ND	ND	No	No	No	ND	ND	Yes	N/A	ND	Tablet

*ND (not disclosed) indicates that the feature was not explicitly described in the publicly accessible documentation reviewed and should not be interpreted as evidence that the feature is absent. N/A indicates not applicable because the system has no corresponding app or portal. No indicates that the system does not have the feature (for example, no app or internet connection). Systems are grouped by neuromodulation type; systems with two types (SCS, Peripheral) are listed under their first type and are counted in both types in Tables 2 and 3. Abbreviations: DBS, deep brain stimulation; HDE, humanitarian device exemption; PMA, premarket approval; PNS, peripheral nerve stimulation; RNS, responsive neurostimulation; SCS, spinal cord stimulation; SNM, sacral neuromodulation; VNS, vagus nerve stimulation.